Implementation of Shrinkage Estimators for Cosmological Precision Matrices

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Objectives

General

Evaluate and extend direct precision shrinkage estimators in the $n \sim N_e$ regime shared by the sequential EnKF and cosmological covariance analysis.

Specific

1. **Implement** four direct-precision targets (identity, eigenvalue, cosmological, and scaled identity) within the TEDA framework.
2. **Benchmark** against Ledoit–Wolf, NERCOME, and the standard EnKF and EnKF-Cholesky baselines on Lorenz-96 and BOSS cosmological mocks.
3. **Characterize** how target structure (diagonal vs. dense) affects the shrinkage coefficients, clamping, and the bias–variance tradeoff.
4. **Propose** a domain-agnostic scaled-identity target as a middle ground between the plain identity and the domain-specific cosmological target.

How the EnKF Works

At each cycle t:

1. **Simulate** N_e parallel forecasts (the ensemble) by propagating the model forward
2. **Estimate** a covariance matrix ($\mathbf{B}_t$) from those forecasts
3. **Invert** it to get a precision matrix ($\mathbf{B}_t^{-1}$)
4. **Use** the precision matrix to blend the forecast with real observations

The precision matrix is **re-estimated and re-used every cycle**.

The Cosmological Parallel

For one analysis:

1. **Simulate** N_{mocks} mock catalogs (synthetic realizations of the universe)
2. **Estimate** a covariance matrix ($\mathbf{\Sigma}$) from those mocks
3. **Invert** it to get a precision matrix ($\mathbf{\Sigma}^{-1}$)
4. **Use** the precision matrix in a Gaussian likelihood to fit parameters against the real measurement

The precision matrix is **estimated once and reused as fixed**.

Same Matrix, Different Names

Step 2 in both workflows computes the same kind of matrix.

	Data Assimilation	Cosmology
Samples	N_e ensemble forecasts	N_{mocks} mock catalogs
Covariance matrix of ...	forecast error	power-spectrum measurement
Name of the matrix	$\mathbf{B}_t$	$\mathbf{\Sigma}$

Same statistical object: a sample covariance matrix estimated from N samples of an n-dimensional vector.

From here on, call it $\mathbf{S}$: the *generic sample covariance matrix*.

The Generic Sample Covariance Matrix

For N samples $\mathbf{x}_1, \ldots, \mathbf{x}_N$:

$$\mathbf{S} = \frac{1}{N-1} \sum_{i=1}^{N} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

Correct on average, but for any single estimate:

- **Very noisy when** $N \sim n$: noise scales as n^2/N
- **Not invertible when** $N \leq n$: the matrix is singular
- **Inversion blows up the noise**: small eigenvalues in $\mathbf{S}$ become huge in $\mathbf{S}^{-1}$

So in the small-N, large-n regime, $\mathbf{S}$ is unusable. We need a better estimator.

Bias vs. Variance

The total error of any estimator splits into two pieces:

$$\text{Total error} \;=\; \text{Bias}^2 \;+\; \text{Variance}$$

- **Bias**: systematic offset from the truth
- **Variance**: random jitter between different sample sets

$\mathbf{S}$ has **zero bias** but **huge variance**.

The idea: accept a small bias in exchange for much less variance.

Shrinkage

Blend $\mathbf{S}$ with a structured target matrix $\mathbf{T}$:

$$\hat{\mathbf{S}} \;=\; (1-\lambda)\,\mathbf{S} \;+\; \lambda\,\mathbf{T}$$

- $\mathbf{T}$: something we trust a priori (e.g., identity, or a domain-informed matrix)
- $\lambda \in [0, 1]$: how much to trust $\mathbf{T}$ vs. $\mathbf{S}$
  - $\lambda = 0$: pure $\mathbf{S}$
  - $\lambda = 1$: pure $\mathbf{T}$

The "better estimators" in the field all differ in **how they choose $\mathbf{T}$ and λ**.

State of the Art

Family	Representative	What it does
Linear covariance	Ledoit–Wolf (2004)	Blends $\mathbf{S}$ with a scaled identity
Target-free	NERCOME (2016)	Averages many bootstrap eigenvalue estimates
Nonlinear	Ledoit–Wolf (2012, 2020)	Uses a different λ for each eigenvalue
Sparse precision	GLasso, CLIME	Forces most entries of $\mathbf{S}^{-1}$ to 0
Direct precision	**Bodnar et al. (2016)**	Blends $\mathbf{S}^{-1}$ directly (our basis)
Cosmological target	Pope–Szapudi (2008), Looijmans (2024)	Uses a domain-specific target matrix
Localization	Gaspari–Cohn; modified Cholesky	Zeroes out long-range correlations in $\mathbf{S}$

BUT EACH HAS A DOMINANT WEAKNESS

Critical Comparison

Four weaknesses across the seven families:

1. **Wrong object.** Most methods shrink $\mathbf{S}$, but we need $\mathbf{S}^{-1}$. Inverting a shrunk $\mathbf{S}$ amplifies small-eigenvalue errors.
2. **Too expensive.** NERCOME bootstraps ~200× per call. Fine once, prohibitive inside a 100-cycle EnKF loop.
3. **Wrong shape.** Diagonal targets fail when the true precision is dense (e.g., Lorenz-96); coefficients explode and clamp every cycle.
4. **Wrong metric.** Methods minimize one-shot error. But the EnKF reuses the estimator every cycle, so errors compound.

The Research Gap

No existing method delivers **all four** at once:

1. Shrinks the **precision** matrix (not just the covariance matrix)
2. **Cheap enough** to run every cycle inside an EnKF
3. **Stable** across sequential cycles (low variance)
4. **Flexible target**: works without domain knowledge

Proposed Method

Start from Bodnar et al. (2016). Instead of shrinking $\mathbf{S}$ and then inverting, **shrink $\mathbf{S}^{-1}$ directly** against a target precision matrix $\mathbf{\Pi}_0$:

$$\mathbf{\Pi}_{\mathrm{LS}} \;=\; \alpha^* \, \mathbf{S}^{-1} \;+\; \beta^* \, \mathbf{\Pi}_0$$

- α^*, β^*: analytical weights, derived so the expected error of $\mathbf{\Pi}_{\mathrm{LS}}$ is minimized (closed-form, recomputed from the data each time)
- $\mathbf{\Pi}_0$: a target precision matrix; our choice

This **fixes weakness #1** from the previous slide: we are now shrinking the object we actually need.

Contribution 1: Sequential Embedding

Embed direct precision shrinkage inside the EnKF loop. At each cycle t:

1. Take the current ensemble
2. Compute $\mathbf{S}$ and recompute α^*, β^* **from scratch**
3. Form $\mathbf{\Pi}_{\mathrm{LS}}$ and feed it into the Kalman gain

No existing paper does this for direct precision shrinkage.
Bodnar et al. (2016) and Looijmans et al. (2024) apply it only to one-shot cosmological estimation.

Four Target Choices Evaluated

Target	$\mathbf{\Pi}_0$	Notes
Identity	$\mathbf{I}$	generic default
Eigenvalue	uses eigenvalues of $\mathbf{S}^{-1}$	adapts to the data's spectrum
Cosmological	$\mathrm{diag}(N_\ell / 2C_\ell^2)$	plugs in domain knowledge
Scaled identity	$\mathrm{diag}(2/\mathbf{S}_{ii})$	**new in this thesis**

One short slide per target follows.

Target: Identity

$$\mathbf{\Pi}_0 = \mathbf{I}$$

- The **generic default**. No assumption about the data
- Treats every variable as if on the same scale
- **Fails** when variables have very different magnitudes (e.g., power spectrum at low vs. high ℓ)

Baseline target in Bodnar et al. (2016); analogue of the scaled identity in Ledoit–Wolf (2004).

Target: Eigenvalue

$$\mathbf{\Pi}_0 = \mathbf{U}\,\mathrm{diag}(\lambda_i)\,\mathbf{U}^T$$

where $\mathbf{U}$, λ_i are the eigenvectors and eigenvalues of $\mathbf{S}^{-1}$.

- **Adapts to the data's own spectrum**. No external parameters
- Keeps the diagonal part of $\mathbf{S}^{-1}$ in its own eigenbasis
- Self-referential: uses $\mathbf{S}^{-1}$ to build its own target

Introduced by Bodnar et al. (2016) as an alternative target for direct precision shrinkage.

Target: Cosmological

$$\mathbf{\Pi}_0 = \mathrm{diag}\left(\frac{N_\ell}{2\,C_\ell^2}\right)$$

- C_ℓ: theoretical power spectrum at multipole ℓ, from cosmological perturbation theory and reference parameters
- $N_\ell \approx (2\ell + 1)\, f_{\mathrm{sky}}$: mode count from survey geometry
- Assumes the underlying density field is **Gaussian**

Requires a fixed cosmology, a specific survey, and a Gaussian approximation.
Doesn't transfer to other domains.

Pope & Szapudi (2008) first used it for covariance shrinkage in cosmology; Looijmans et al. (2024) applied it with Bodnar's direct precision framework.

A Gap in the Target Menu

Each existing target has a weakness:

- **Identity**: ignores the scale of each variable
- **Eigenvalue**: **self-referential** and **operationally heavy** (uses $\mathbf{S}^{-1}$ to build its own target; doubles per-cycle cost)
- **Cosmological**: **requires domain knowledge** (a specific cosmology and survey)

Missing: a target that is **scale-aware**, **simple**, and **domain-agnostic**.
Built from a cheap data summary, not from the full noisy inverse.

Contribution 2: Scaled Identity Target

$$(\mathbf{\Pi}_0)_{ii} = \frac{2}{\mathbf{S}_{ii}}$$

- **Scale-aware**: per-variable variances from the diagonal of $\mathbf{S}$ itself
- **Domain-agnostic**: no external parameters, no cosmology, no survey
- **Simple**: only reads the diagonal of $\mathbf{S}$; $\mathcal{O}(n)$ per cycle
- Factor of 2 mirrors the cosmological $2/N_\ell$ scaling

Fills the gap: scale-aware + domain-agnostic + simple.

Why It Works in Practice

Consequences of the scale-aware + domain-agnostic + simple design:

- **Magnitude-matched target**: diagonal $2/\mathbf{S}_{ii}$ is already in the right order of magnitude as $\mathbf{S}^{-1}$, so α^* and β^* stay in $[0, 1]$ far more often than with plain identity → **lower clamping rate**.
- **Data-driven but cheap**: unlike the eigenvalue target (doubles per-cycle cost) or cosmological (needs pre-analysis setup), it's a one-line diagonal read.

Result: **10–30% improvement** over plain identity on both testbeds (RMSE in DA, Frobenius loss in cosmology).

Testbed 1: Lorenz-96

A chaotic toy model of atmospheric dynamics. State dimension $n = 40$.

- Forcing $F = 8$ (strongly chaotic), 4th-order Runge-Kutta integration
- Observations: 32 of 40 variables observed, Gaussian noise $\sigma_o = 0.01$
- Inflation factor $\rho = 1.05$
- Assimilation window: 100 cycles
- Repetitions: **30 runs** per configuration

Neighboring variables are coupled through the dynamics → **true precision matrix is dense off-diagonal**. The hard case for diagonal targets.

Testbed 2: AR(1) Cosmological Model

A simplified dynamical model whose stationary statistics match BOSS DR12 cosmological mocks. State dimension $d = 18$ (9 wavenumbers × 2 multipoles).

- Persistence $\phi = 0.95$ (smooth time evolution), calibrated to 2048 Patchy mocks
- Stationary covariance equals the BOSS mock covariance by construction
- Observations: 14 of 18 variables observed, Gaussian noise $\sigma_o = 100$
- Inflation factor $\rho = 1.02$
- Repetitions: **30 runs** per configuration

Cosmological modes are nearly independent → **true precision matrix is near-diagonal**.
The easy case for diagonal targets.

Why Both Testbeds?

The two systems have **opposite precision structure** by design:

- **Lorenz-96**: dense off-diagonal precision (hard case)
- **AR(1) cosmological**: near-diagonal precision (easy case)

Running the **same methods** on both lets us separate two effects:

- Does the **method** itself work?
- Does the **target match the truth**?

A single testbed would conflate the two.

Eight Methods Compared

Head-to-head on both testbeds:

- **Baselines**: EnKF, EnKF-Cholesky
- **Covariance shrinkage**: Ledoit–Wolf
- **Target-free**: NERCOME
- **Direct precision shrinkage** (our framework), with four targets:
  - Identity
  - Eigenvalue
  - Cosmological *
  - **Scaled identity** (new in this thesis)

* The cosmological target can't be built on Lorenz-96 since it relies on domain-specific cosmological quantities. Evaluated only on the cosmological testbed.

Ensemble Size: Lower Bound

How small can N_e be? Two constraints:

- $N_e > n + 2$: needed for $\mathbf{S}$ to be invertible
- $N_e > n + 4$: needed for the Bodnar coefficients (α^*, β^*) to be well-defined

For Lorenz-96 ($n = 40$), that's $N_e \geq 44$. Our smallest tested value is $N_e = 50$, six members above the bound.

Anything smaller falls into the **under-sampled regime** that this thesis doesn't cover.

Ensemble Size: Upper Bound

How large can N_e be before shrinkage becomes pointless?

As $N_e \gg n$:

- The noise in $\mathbf{S}$ shrinks (scales as n^2/N_e)
- The shrinkage weight λ drops toward 0
- $\mathbf{S}$ is already good, the target stops mattering

Beyond $N_e \approx 3n$, shrinkage offers no meaningful gain over plain $\mathbf{S}$.

For Lorenz-96 we stop at $N_e = 100$ ($n/N_e = 0.4$), safely inside the interesting regime.

Tested Ensemble Sizes: Lorenz-96

State dimension $n = 40$.

N_e	n/N_e	Regime & role
50	0.80	Ill-conditioned. Stress at the invertibility bound
60	0.67	Moderately ill-conditioned
80	0.50	Moderately ill-conditioned
100	0.40	Moderate → well-sampled boundary

Four values chosen to span the regime between the lower and upper bounds.

Tested Ensemble Sizes: Cosmological

State dimension $d = 18$.

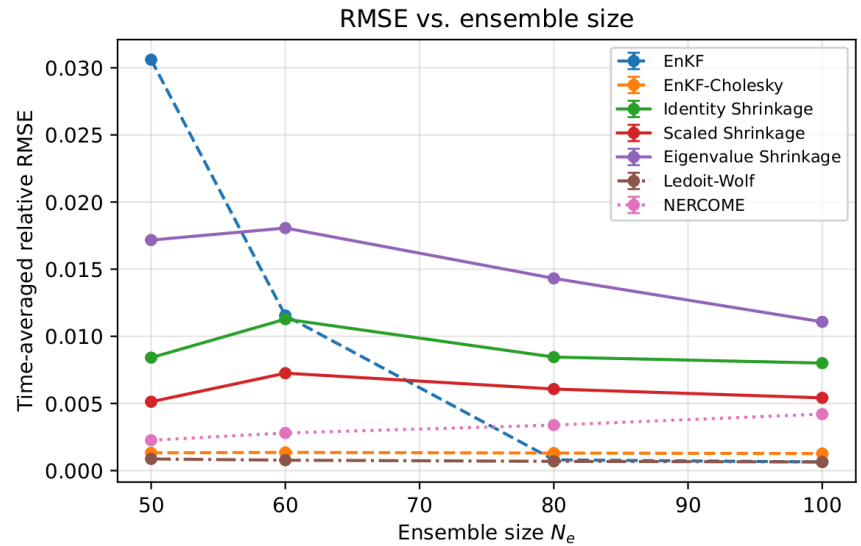
N_e	n/N_e	Regime & role
24	0.75	Ill-conditioned. Matches operational mock budgets
30	0.60	Moderate (Looijmans et al. working point)
40	0.45	Moderate
50	0.36	Moderate → well-sampled boundary

Validated empirically with a finer 12-value N_e scan (in the appendix).
No sharp transitions between these points.

RMSE vs N_e on Lorenz-96

- Direct precision methods: **erratic**, RMSE 0.5–1.8 × 10^{-2}
- Ledoit–Wolf: **lowest** RMSE, stable across N_e
- EnKF-Cholesky: competitive via banded structure
- Standard EnKF: diverges at small N_e

Cause: dense precision + diagonal targets → severe coefficient miscalibration.



Why Do Direct Precision Methods Fail on Lorenz-96?

Look at the shrinkage coefficients:

- α^* stays within its valid range, near the theoretical bound $1 - n/N_e$.
- β^* **blows up to** 10^8**–**10^9, far outside the valid range $[0, 1]$, so it gets clamped back to 1.
- **Clamping happens > 80% of cycles** across all three diagonal targets.

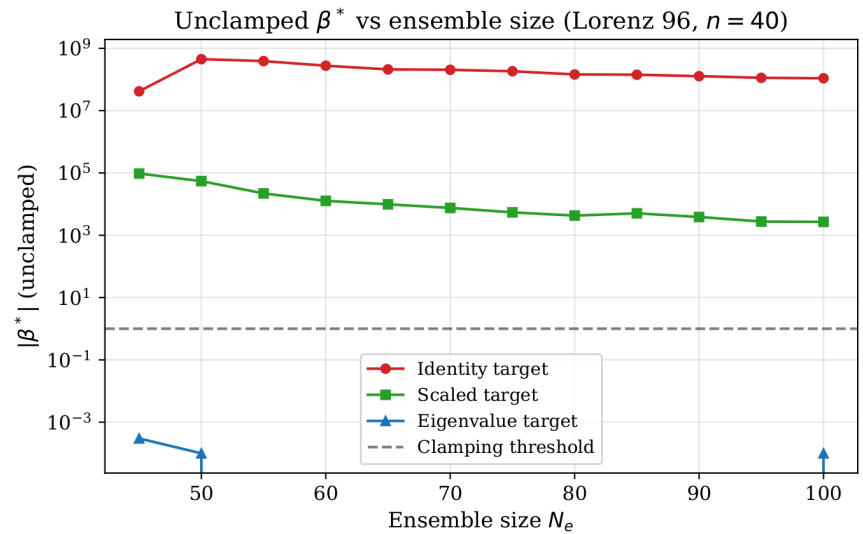
Result: the precision matrix $\mathbf{\Pi}_{\mathrm{LS}}$ is highly ill-conditioned (condition number 10^7–10^{11}, vs. 10^2–10^3 for Ledoit–Wolf).

Root cause: dense precision + diagonal target = forced mismatch. The Bodnar formula tries to correct by inflating β^*; clamping prevents the correction.

Unclamped β^* Explodes

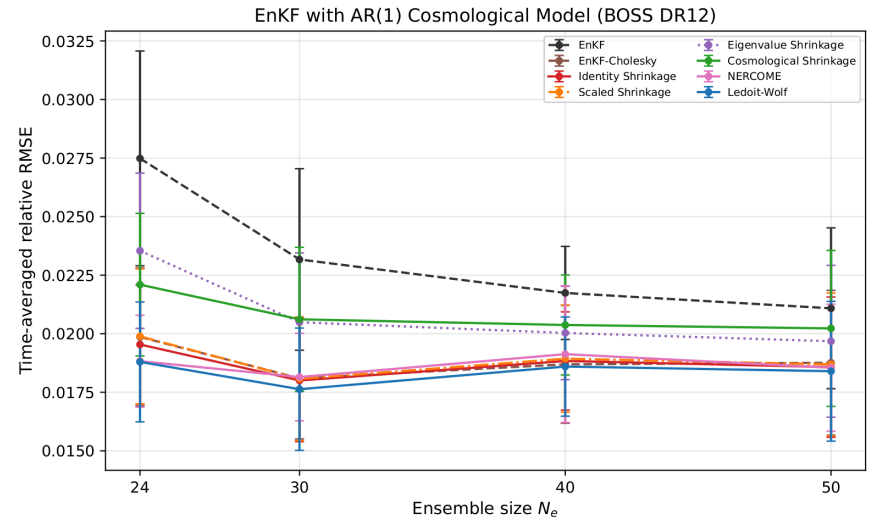
Unclamped $|\beta^*|$ vs N_e (log scale). Dashed line: clamp threshold at $\beta^* = 1$ (valid range $[0, 1]$).

- **Identity target:** 10^7–10^9 times too large
- **Scaled identity:** better, but still needs clamping
- **Eigenvalue target:** the only one that behaves normally here



Cosmological Domain

- All methods stable, **monotonic RMSE reduction** with N_e
- Direct precision with cosmological target → lowest Frobenius loss
- Scaled identity: second-best domain-agnostic method
- **Ledoit–Wolf wins in EnKF RMSE**



The Central Finding

Best standalone estimator ≠ best EnKF filter.

A disconnect not previously documented in the shrinkage literature.

Standalone vs Sequential (Cosmological Domain)

Standalone precision estimation (one-shot Frobenius loss):

- Direct precision with **cosmological target** wins, loss ≈ 0.47
- Other direct precision targets: eigenvalue > scaled identity > identity (all ≈ 0.54)
- Ledoit–Wolf worst, loss ≈ 0.93–0.96

EnKF filtering (RMSE over 100 cycles):

- Ledoit–Wolf **wins**
- Direct precision with cosmological target: second
- Other direct precision targets: scaled identity > eigenvalue > identity
  (ours is best among domain-agnostic)

Why the Ranking Flips

The EnKF applies the estimator at every cycle, so **variance compounds**.

- Ledoit–Wolf std across draws: ± 0.01
- Direct precision std: ± 0.13–0.14 (order of magnitude larger)

Over 100 cycles, a 100-fold variance gap overwhelms the bias advantage.

Principle: low-variance beats low-bias when errors accumulate sequentially.

Asymptotic Cost

Method	Cost per cycle
EnKF	$\mathcal{O}(n^2 N_e + m^2 N_e)$
EnKF-Cholesky	$\mathcal{O}(n\, r\, N_e)$
Direct precision (identity / scaled identity / cosmo)	$\mathcal{O}(n^2 N_e + n^3)$
Direct precision (eigenvalue target)	$\mathcal{O}(n^2 N_e + 2n^3)$
NERCOME	$\mathcal{O}(B(n^2 N_e + n^3))$
Ledoit–Wolf	$\mathcal{O}(n^2 N_e)$

n: state dimension · N_e: ensemble size · m: observations · r: Cholesky bandwidth · B: bootstrap repeats (~200)

Cost Takeaways

- **Direct precision**: adds an $\mathcal{O}(n^3)$ inversion on top of what EnKF already does
- **NERCOME**: multiplies the cost of direct precision by B (~200 bootstrap repeats)
- **EnKF-Cholesky**: the only method **linear in** n; only one that scales to operational dimensions

Measured Wall-Clock

Lorenz-96 ($n = 40$, $N_e = 80$, per 100 cycles):

- All shrinkage methods: 26–28 s
- NERCOME: **215 s**, 8× slower

Cosmological ($d = 18$, $N_e = 30$, per 100 cycles):

- Shrinkage methods: 0.20–0.28 s
- NERCOME: **26.7 s**, 190× slower
- EnKF baseline: 0.14 s

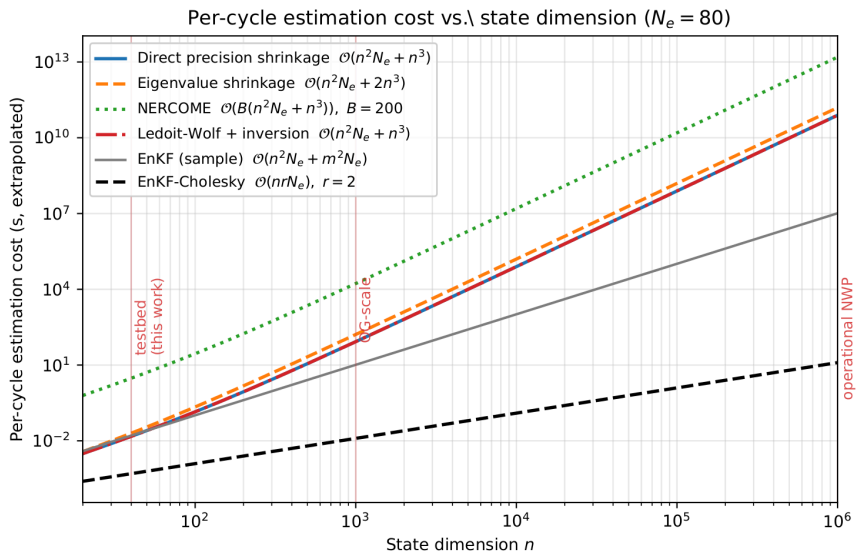
Ensemble propagation dominates on Lorenz-96; precision estimation dominates on cosmo.

Scaling to Operational Dimensions

Extrapolation from measured timings using asymptotic formulas:

- $n \lesssim 100$: cost not a differentiator
- $n \sim 10^2$–10^3: n^3 inversion dominates; NERCOME ruled out
- $n \gtrsim 10^4$: only EnKF-Cholesky (linear in n) survives

Practical implication: direct precision shrinkage is best suited to mid-scale testbeds or combined with localization at operational scale.



Practical Recommendation

Regime (n/N_e)	Precision structure	Target known?	Recommended
> 1 (under-sampled)	any	n/a	Localization + EnKF-Cholesky
0.5–1	near-diagonal	**yes**	Direct precision + domain target
0.5–1	near-diagonal	**no**	**Scaled identity** (this work) or Ledoit–Wolf
0.5–1	dense	n/a	Ledoit–Wolf; EnKF-Cholesky at $n \gtrsim 10^3$
< 0.3 (well-sampled)	any	n/a	Plain EnKF (shrinkage weight $\lambda \to 0$)

Accuracy drives the choice within the $\mathcal{O}(n^3)$ band; cost drives it at the extremes.

Contributions

1. **First sequential embedding** of Bodnar direct precision shrinkage, coefficients recomputed every cycle inside the EnKF.
2. **Novel scaled identity target**, a data-driven middle ground between the plain identity and the domain-specific cosmological target.
3. **Dual-domain benchmark** on systems with opposite precision structure, separating method design from target-truth alignment.
4. **Standalone ≠ sequential**, the key finding: estimator variance matters as much as bias when errors accumulate.
5. **Systematic ensemble-size study**, documenting when and how the shrinkage coefficients fail.
6. **Open-source implementation** in the TEDA framework for all six estimators.

Limitations

- **Under-sampled regime not covered**, when $N_e < n$, $\mathbf{S}$ is singular and all direct precision methods break. Would need localization.
- **Low-dimensional testbeds**, $n = 40$ and $d = 18$ are tiny compared to operational DA (10^6–10^8).
- **Simplified cosmological model**, AR(1) is linear; real surveys involve nonlinear structure growth.
- **Linear observation operators only**, no tests with nonlinear measurement models.
- **Heuristic factor of 2** in the scaled target, picked by analogy to cosmology; could be cross-validated instead.

Future Work: Methodology

- **Shrinkage × localization**, combine the two regularization strategies
- **Intermediate-scale systems**, test scaling on quasi-geostrophic models ($n \sim 10^4$)
- **Adaptive target selection**, switch target based on the data's off-diagonal structure
- **Theoretical analysis of sequential accumulation**, derive bias/variance trade-off rigorously

Future Work: Estimator Extensions

- **Principled scaling factor**, replace the heuristic 2 with a cross-validated value
- **Non-Gaussian samples**, extend Bodnar's derivation beyond Gaussian ensembles
- **Condition-number bound**, truncate tiny eigenvalues to prevent inversion blow-up
- **Time-adaptive target**, update the target from a running mean of past estimates

Thank you

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